TryHackMe_Pyramid_of_Pain.md

The Pyramid of Pain framework ranks Indicators of Compromise (IoCs) based on the level of "pain" or effort caused to an adversary when defensive teams detect and block those specific indicators.

Task 2: Hash Values (Trivial)

Concept: Cryptographic file hashes (MD5, SHA-1, SHA-256) uniquely identify specific malware samples. Blocking a hash causes minimal disruption (trivial pain) to an adversary because changing even a single byte in a file completely alters its hash.

Questions & Answers:

Analyse the report associated with the hash b8ef959a9176aef07fdca8705254a163b50b49a17217a4ff0107487f59d4a35d. What is the filename of the sample?

Answer: Sample.exe (or vserv.exe / file name displayed in the provided report)

Task 3: IP Address (Easy)

Concept: IP addresses identify attacker infrastructure (e.g., C2 servers). Blocking an IP address causes slight disruption ("easy" pain), but attackers can quickly bypass IP blocks using dynamic DNS or Fast Flux networks.

Questions & Answers:

What is the first IP address the malicious process (PID 1632) attempts to communicate with?

Answer: 50.87.136.52

What is the first domain name the malicious process (PID 1632) attempts to communicate with?

Answer: craftingtheweb.com (or domain linked in the any.run sample report)

Task 4: Domain Names (Simple)

Concept: Domain names require registration and payment. Blocking domains forces the attacker to buy or register new ones, which takes time.

Questions & Answers:

Go to this report on app.any.run and provide the first suspicious domain request you are seeing...

Answer: craftingtheweb.com

What term refers to an address used to access websites?
Answer: Domain Name
What type of attack uses Unicode characters in the domain name to imitate a known domain?
Answer: IDN Homograph (or Homograph attack)
Provide the redirected website for the shortened URL using a preview:
https://tinyurl.com/bw7t8p4u
Answer: https://tryhackme.com/

Task 5: Host Artifacts (Annoying)

Concept: Host artifacts are traces left by attackers on an endpoint (e.g., specific registry keys, dropped files, custom process execution paths). Detecting these forces the adversary to re-engineer their deployment routines.

Questions & Answers:

A process named regidle.exe makes a POST request to an IP address based in the United States (US) on port 8080. What is the IP address?
Answer: 96.126.101.6
The actor drops a malicious executable (EXE). What is the name of this executable?
Answer: G_Jupk.exe
Look at this report by VirusTotal. How many vendors determine this host to be malicious?
Answer: 9 (or the vendor count from the specified VT report link)

Task 6: Network Artifacts (Annoying)

Concept: Network artifacts include HTTP User-Agent strings, custom URI request patterns, and C2 communication protocols observed in PCAP files.

Questions & Answers:

Which network indicator helped us identify the malware type (Emotet)?
Answer: User-Agent
How many POST requests are in the screenshot from the PCAP file?
Answer: 6

Task 7: Tools

Concept: Blocking software utilities (like Cobalt Strike, Mimikatz, or custom stealers) forces attackers to rebuild or obfuscate their entire toolset. Detection rules often rely on YARA rules or fuzzy hashing (such as SSDeep).

Questions & Answers:

Provide the method used to determine similarity between the files:
Answer: Fuzzy Hashing
Provide the alternative name for fuzzy hashes without the abbreviation:
Answer: Context Triggered Piecewise Hashing

Task 8: TTPs

Concept: Tactics, Techniques, and Procedures (TTPs) encompass the attacker's operational behavior mapped to the MITRE ATT&CK Matrix. Detecting TTPs causes maximum "pain," as it invalidates the adversary's entire attack workflow.

Questions & Answers:

Navigate to ATT&CK Matrix webpage. How many techniques fall under the Exfiltration category?
Answer: 9
Chimera is a China-based hacking group... What is the name of the commercial, remote access tool they use for C2 beacons and data exfiltration?
Answer: Cobalt Strike

Task 9: Practical: The Pyramid of Pain

Concept: Sorting indicators into their correct tiers within the interactive Pyramid of Pain simulation.